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SURVEY

Generative Adversarial Networks Assist Missing Data Imputation: A Comprehensive Survey and Evaluation

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ABSTRACT Missing data imputation is a technique to deal with incomplete datasets. Since many models and algorithms cannot be applied to data containing missing values, a pre-processing step needs to be performed to remove incomplete data or to estimate the missing values. This is a well-known problem referred to as the *data imputation problem*. Several approaches have been designed for data imputation. These algorithms can be divided into two main categories: statistical and machine learning-based algorithms. As machine learning algorithms are optimized, they usually have better performance compared with statistical ones. In this paper, we review the most recent literature related to missing data imputation based on generative adversarial networks (GANs) that have gained tremendous attention in dealing with missing values. We examine the structures of GANs for missing data imputation and discuss the commonly used datasets and metrics for evaluation. We also cover the influence of the missing datatype, the effect of the missing data fraction, and the algorithm-related problems on data imputation performance. We conduct experiments on two publicly available datasets and evaluate the performance of *GAIN*, a missing data imputation algorithm to that of existing state-of-the-art approaches, demonstrating that the GAN-based algorithm outperforms the others in terms of RMSE and FID.

INDEX TERMS Generative adversarial networks, missing data, data imputation, semi-supervised learning, data cleaning.

I. INTRODUCTION

Missing data is one of the major challenges for statistical analyses of many different research fields. Many machine learning models can not directly handle datasets with missing information. Different reasons can lead to missing values including but not limited to incomplete data entry, equipment malfunctions, lost files, cyber-attack, security, and privacy concerns, and many other reasons [1], [2], [3]. For example, the unwillingness of people to fill some of the required values during the data collection phase [4], missing values in clinical analyses of patients [5], and sensor damages [6], [7]. The removal of rows or columns of incomplete data in the database is a simple but inefficient solution to this problem. However, this can significantly reduce the number of samples, especially if the dataset contains numerous samples with missing values [8].

In real-world problems, the datasets are usually imperfect and contain many samples with missing values. For instance in traffic data [9], smart grids [10], and electronic health record (EHR) data [11]. In this case, removing the samples containing incomplete data damages the effectiveness of algorithms. For instance, considering data collected by a hospital information system, for instance, EHR, we have data collected by medical experts for specific purposes and data collected by the information system without any specific purpose, that might be useful in future research. Often, the second type of collected data contains many instances with missing values [5], whereas, in some other applications, this missing data could be caused by confidential data are not

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available [12]. On the other side, sometimes missing data may have essential information that should not be neglected. For instance, in economic transactions values regarding the transferred cash, cannot be missing, especially if the expected quantity could be very large. All these motivate the researchers to study the missing data imputation problem and propose the proper methods to handle these missing data.

Missing data imputation is a well-studied subject. There are many studies that address the related imputation algorithms. We generally divide these methods into two main categories: i) statistical and ii) machine learning-based algorithms. For statistical algorithms, the aim is to impute the missing values by a handy designed algorithm (e.g., mean or mode), whereas for machine learning algorithms the aim is to optimize a function taking in input several parameters, including the type of data. Designing a statistical model to deal with different kinds of missing values (with a different percentage of these values) is much more difficult compared with machine learning algorithms. This is because machine learning algorithms follow an optimization process. Besides the flexibility of machine learning algorithms, they can extract more complex patterns of data [13]. Therefore, the investigation of imputation-based machine learning algorithms is necessary to choose an appropriate strategy for the desired task.

Among the machine learning algorithms, the generative adversarial network (GAN) [14], [15] has attracted great attention due to its capability for generating synthetic data. Researchers have introduced many evolved variants of GAN, including, but not limited to, conditional GAN [16] for generating the class labels, Wasserstein GAN [17] to stabilize the model during the training phase, InfoGAN [18] to use additional information besides the inputs, and least-squares GAN [19] for generating high-quality images. All these GAN-based algorithms have been designed for generating synthetic samples. Recently, GAN-based algorithms have been developed for missing value imputation. In this paper, we explore GAN-based missing data imputation and discuss existing approaches in terms of their applicability and frameworks. We present some of the literature's available and commonly used benchmark datasets and discuss the effectiveness of GAN-based algorithms in terms of their missing technique. To properly address this paper's contributions, we first examine existing comparable surveys and present our contributions in Section I-B.

A. RELATED EXISTING SURVEYS

Missing data imputation is an important and well-studied subject. Wei-Chao Lin and Chih-Fong Tsai studied the basics and literature of the data imputation algorithms for statistical and machine learning algorithms covering the studies published 2006-2017 in [12]. Their research includes mostly the common machine learning (ML)-based algorithms without addressing the technical aspects and evaluations.

Anil Jadhav et al. compare seven basic data imputation algorithms on five different numeric datasets [20]. They compare mean, median, K-nearest neighbor (KNN), predictive mean matching, Bayesian Linear Regression (norm), Linear Regression, non-Bayesian, and random samples on datasets obtained from the UC Irvine machine learning repository[1] by using root mean square error (RMSE) as the evaluation metric. In their comparisons, the KNN outperforms the other basic algorithms. The authors do not address the GANs.

The authors in [21] review the autoencoder-based data imputation algorithms. They study the imputation of tabular data with autoencoders, for the studies published during 2014-2020. The survey reviews both datasets and evaluation metrics.

Jaeyoon Kim et al. study GAN-based data imputation. They mainly present the block diagram of the original GAN and very shortly, in three pages, present some of the existing algorithms just by mentioning their names. They do not provide the technical details, structure presentation, discussions on the advantages or disadvantages, datasets, evaluation metrics, and experimental results [22].

In a recent survey by Miao et al. [23] the authors conduct many experiments to evaluate ML-based missing data imputation algorithms. They employ the average mean absolute error (AMAE) and average root mean square error (ARMSE) as evaluation metrics, [24], and run simulations on many datasets from the UCI machine learning repository. In their evaluations, they use GAIN and VGAIN as GAN-based algorithms. Their evaluation results further demonstrate that GAN-based algorithms are successful and stable in missing data imputations. They do not, however, examine the structure or the numerous variants of GAN-based algorithms in the current literature.

Some existing surveys concentrate on applications rather than algorithms. Yuan Luo extracted 13 measured blood tests, eliminating one recorded result from each laboratory at random. He demonstrates that ML-based and statistical models, like LightGBM and multiple imputation by center equation (MICE), can achieve high imputation performance. [25]. Yifan Zhang et al. investigate missing data imputation in environmental monitoring, taking water quality factors and imputation methods into account. They claim that ML-based recurrent neural networks (RNNs) can handle data in near real-time [26].

Ardalan Mirzaei et al. study the surveys of missing data imputation and review some of the handling methods including the data deletion and likelihood. They do not study GANs in missing data imputation [27]. Kimset Seo et al. review the ML-based data imputation methods including the KNN, bayesian principal component analysis (BPCA), MICE, and multiple imputations with denoising autoencoder neural network (MIDAS). They also review the mean absolute error (MAE) and RMSE metrics and show that the

[1]The datasets are publicly accessible at https://archive.ics.uci.edu/datasetshttps://archive.ics.uci.edu/datasets

KNN and MICE outperform the BPCA and MIDAS data imputation algorithms [28].

Some review articles focus on the specific application, for example, missing data imputation on water distribution system [29], cardiac imaging data [30], clinical trials [31], network analysis [32], Internet of things (IoT) [33], and clinical epidemiological research [34]. Also, different surveys have covered the GAN-based algorithms for generating synthetic samples [35], [36], [37], [38], [39], [40].

Existing surveys, according to our findings, do not cover GAN-based data imputation techniques. The majority of the surveys deal on ML-based data imputation, and the only two papers that cover GAN do so insufficiently. This encouraged us to study and investigate existing cutting-edge algorithms, providing a technical overview in terms of datasets used and evaluation criteria, as well as some experimental results comparing traditional ML-based missing data imputation approaches with GANs. Unlike prior surveys, this work focuses on GAN-based methods rather than specific applications. Table 1 compares our work with other related state-of-the-art surveys.

B. OUR CONTRIBUTIONS

To the best of our knowledge, there is no comprehensive survey for GAN-based missing data imputation. The goal of this study is to investigate the generative adversarial networks that are used for missing data imputation. We also briefly cover ML-based data imputation strategies in order to properly address GANs. We explore the impact of missing data kinds, missing data fractions, and algorithm-related challenges on data imputation performance. We provide and compare metrics and datasets used in missing data imputation studies. The key contributions of this survey are as follows:

- We concentrate on a thorough examination of GAN-based data imputation techniques and present the algorithm, network structure, and applications of several GANs. We categorize these algorithms into regular, time series, pictures, and specific purpose categories. We also discuss statistics and other ML-based techniques briefly.
- We look at publicly available datasets and evaluation metrics that are frequently used to evaluate data imputation techniques. We provide access links as well as metrics formulations and dataset descriptions.
- We address the advantages and drawbacks of GAN-based data imputation. We evaluate the efficacy of numerous GAN-based algorithms in dealing with different types and fraction rates of missing data. In addition to the current issues, we discuss possible prospective opportunities.
- We implement some existing imputation algorithms and compare GAN-based imputation to other cutting-edge techniques. Our findings show that the GAN-based imputation algorithm significantly improves imputation accuracy in terms of RMSE and Frechet Inception distance (FID).

The remainder of this work is structured as follows: We discuss data imputation strategies briefly in Section II. Section III discusses the reviewed GAN-based imputation algorithms, guiding readers through the process of developing a GAN-based imputation architecture. The evaluation metrics, datasets, and open-source scripts are studied in Section IV. Section V discusses the current obstacles and research opportunities in GAN-based data imputation techniques. Finally, Section VI finalizes the paper and discusses future works.

II. DATA IMPUTATION METHODS

Imputation algorithms are constantly improving, and researchers have developed a variety of ways for dealing with missing data [42]. This section provides an overview of statistical and machine-learning approaches for the imputation of missing values in incomplete datasets.

A. STATISTICAL ALGORITHMS

The statistical algorithms are related to the statistical perspective of data and do not have any optimization process. For instance, mean [43], median [44], and mode [45] are three of the well-known statistical techniques [12]. The mean and median techniques are appropriate for imputing numerical attributes, and the mode technique is suitable for categorical ones. In the mean and median approaches, missing values of an attribute are replaced with the average and median values of that attribute in the observed (complete) data. On the other hand, in the mode approach, the missing values of an attribute are replaced by a value, which is the most common value on that attribute. Also, there are some other techniques such as principal component analysis (PCA) [46] and singular value decomposition (SVD) [47] that project the data into another space, where the attributes do not have the same value, and the projected data are re-constructible from the projected space. Since the goal of this paper is not to review statistical techniques, we skip further discussions.

B. MACHINE LEARNING ALGORITHMS

The main difference between machine learning algorithms and statistical ones is that the former follows an optimization process. Several machine learning algorithms for data imputation have been proposed so far. For instance, clustering algorithms [48], [49], [50], [51], k-nearest neighbors (KNN) [52], [53], [54], [55], [56], Gaussian process regression (GPR) [57], [58], support vector machine (SVM) [59], [60], long short-term memory (LSTM) [61], [62], [63], [64], [65], decision trees (DT) [66], random forests (RF) [67], [68], [69], auto-encoder (AE) [42], [70], [71], [72], [73], [74], [75], and Generative adversarial networks (GANs). The focus of this paper is to investigate GAN-based models, which will be explained in Section III. The other machine learning algorithms are briefly discussed in the following.

Clustering based algorithms are usually unsupervised learning techniques, that try to group the samples with

TABLE 1. Comparison of this survey with existing related review studies in the field of missing values imputation.

Publication	Summary	Covers Algorithms	Covers GANs Comparisons	Metrics Comparisons	Dataset Description
Lin et al. (2020) [12]	A Comprehensive study of basic missing value imputation techniques	yes	-	yes	yes
Platias et al. (2020) [8]	Comparison of basic machine learning algorithms for data imputation	yes	-	-	yes
Garciarena et al. (2017) [41]	A study on the effect of data imputation algorithms for classification problems	yes	-	-	yes
Jaeyoon Kim et al. (2020) [22]	A very brief introduction to GAN-based algorithms	-	yes (only the basic GAN)	-	-
Ricardo et al. (2020) [21]	Review of Autoencoders for Missing Data Imputation	yes	-	-	yes
Jadhav et al. (2019) [20]	Comparing the basic data imputation algorithms	-	-	-	-
Ardalan et al. (2022) [27]	Missing data in surveys: Key concepts, approaches, and applications	yes	-	-	-
Kimset et al. (2022) [28]	Review of Missing data imputation techniques	yes	-	-	yes
Miao et al. (2023) [23]	Experimental Review of Missing data imputation algorithms	yes	yes (only GAIN, VGAIN)	yes (3 metrics)	yes
Our Work	A comprehensive study of data imputation-based machine learning algorithms focused on GAN-based methods	yes	yes (20 GANs)	yes	yes

similar characters. The centroid of each cluster is the mean of the samples in that cluster. In order to impute the missing values for a given sample, the distance between that sample and the centroid of clusters is calculated. Then, the ''closest'' cluster is chosen and its centroid is chosen to impute the missing values. The most important challenge of clustering-based approaches is the way of splitting the clusters. Based on the splitting strategy, they have different complexities and precision. As an example, the hierarchical method is a well-known strategy to handle the splitting task. At the beginning of clustering, it assumes that each observation stands for a separate cluster, then, two adjacent observations are merged for creating new clusters. The process is continuously repeated till reaching one cluster. A similar splitting task also exists, which uses a reverse process. Authors in [76] have well-studied the most common clustering-based approaches. Clustering-based approaches for data imputation work well when there exist dissimilarities in the data, but they do not work properly in the presence of noise and outliers [77].

k-Nearest Neighbors (KNN) is a well-known supervised learning algorithm, which can also be used for data imputation. First, the distance between a given sample and available samples is calculated, and then, the missing values are imputed by the mean (or weighted mean) of the k nearest samples. The Euclidean distance is usually used to calculate the distance between the samples. There are different types of KNN-based imputation algorithms such as SKNN [78], IKNN [79], CKNN [80] and ICKNN [52], and based on calculating the distances, weighting strategies, and choosing the k value, they have different performances and complexities. They may be useful for some applications and inappropriate for others. For instance, CKNN does not work properly in the case of a high missing rate. However, ICKNN works better than CKNN when the missingness rate is high. For large datasets, KNN-based methods usually implicate huge calculations for measuring the distances and finding k nearest samples.

Gaussian Process Regression (GPR) is a non-linear probabilistic machine learning algorithm, which can also predict the variance of outputs. GPR-based imputation algorithms usually use posterior distribution based on a prior assumption to impute the missing values. GPR is a probabilistic algorithm and therefore, estimates a probabilistic region for the missing values rather than a point estimation. GPR-based algorithms usually utilize kernel functions such as radial basis function, Matérn, Rational quadratic, Exp-Sine-Squared, and linear. Different kernels have different characteristics and are utilized for extracting different attributes from a set of data. For instance, some kernels (e.g., Exp-Sine-Squared) are utilized when the nature of data is periodic, and some kernels (e.g., linear) are used for extracting linear characteristics of data. Some data have a combination of characteristics, and we can model them with a combination of kernels. For instance, Exp-Sine-Squared+linear is utilized when data have both linear and periodic natures. Different GPR-based algorithms have been introduced for imputing the missing values such as NGPR [81] and MTGPR [82] for different applications. One important drawback of GPR-based algorithms is their high computation complexity which makes them unsuitable for large datasets.

Support Vector Machine (SVM) is one of the well-known machine learning algorithms, which can be employed for both classification and regression tasks. Linear regression is usually utilized to separate the samples of classes when the classes are linearly separable. However, in real-world

problems, very often the samples cannot be linearly separated. Therefore, a trick is used for SVM in which the input space is mapped into a high-dimensional feature space modeling a non-linear behavior. Similar to GPR, the SVM uses different types of kernels with different characteristics for converting the data into the feature space. SVM works well when dissimilarities exist between the data and is unsuitable for large datasets due to its time-consuming complexity.

Decision Tree (DT) builds a tree in which the input dataset is partitioned into groups of samples. Each group is characterized by a set of patterns which are the paths from the root to the leaves associated with the group of samples. Data imputation can be performed by considering the samples associated with the same leave of the incomplete sample. There are different versions of DTs such as ID3, C4.5, CRAT, CHAILD, and QUEST [83], [84]. They use different strategies to build trees. Some DTs use two phases of *growing* and *pruning* for constructing the trees (e.g., CRAT and C4.5), and others only use the growing phase. Some versions of DTs are only used for discrete values of targets (e.g., ID3 and C4.5) and cannot handle continuous ones. DTs are non-parametric algorithms and which means that there is no prior assumption about the distribution of data during the training phase. Since a simple DT cannot extract enough information from the data, a group of DTs can help for achieving this goal. The *Random Forest* (RF), which consists of multiple DTs, is usually built according to the bootstrapping process, in which the average of DTs' predictions is considered as the final estimation. The imputation-based DT/RF performs similarly to the KNN process, in which a sample containing the missing value is imputed by the sample of the outcome leaves.

Expectation Maximization (EM) is utilized to obtain a local maximum likelihood parameter of a statistical model where they cannot be obtained directly. These models entail latent variables besides the observed data and unknown parameters. In the context of data imputation, these latent variables can be missing values in the data. Obtaining the maximum likelihood usually needs the likelihood function's derivatives regarding all the unknown values, parameters, and latent variables. Besides, we need to solve the equations, simultaneously. Solving these equations is typically impossible in statistical models. Instead of solving these equations, the EM algorithm, which consists of two numerical steps called expectation and maximization, is utilized to find these unknown values. The EM algorithm guarantees that the likelihood will increase. However, it has a slow convergence [85].

Auto-Encoder (AE) is a class of deep learning algorithms, which learns a coded vector from the input space at the hidden layer in an unsupervised manner. The AE generally consists of three layers: the input layer, the hidden layer, and the output layer. The input samples are used in both the input and output layers, and the goal is to map the input layer to the hidden layer and then reconstruct the input samples through the hidden vector. After the training phase is finished, the input samples containing missing values are fed to the input layer, and then they are reconstructed through the hidden layer. For imputation purposes, the elements of input attributes can be missed randomly in the training phase. Therefore, it is expected that the vectors in the input space consist of randomly missing, and the output layer has the complete set of vectors. Performing this task, the AE will learn how to complete the missing data. Different versions of AEs have been introduced, such as VAE, DAE, SAE, and SDAE. They are used for different purposes. For instance, VAEs are generative models which can produce new samples for imputation in a normal situation, while DAEs are mostly useful in the presence of noise.

Long Short-Term Memory (LSTM) is a deep learning algorithm, which is applicable for time-series problems [86], such as traffic data [87], air quality data [88], and building energy data [61], where data streams are related to time series. LSTM is a type of recurrent neural network, which outperforms conventional machine learning algorithms for this kind of data due to its special structure for extracting time-series information. They take long and short-term information into account during the training process. We cannot state its pros and cons for imputing purposes in comparison with previous machine learning algorithms due to its application for time series data. However, LSTMs suffer from a vanishing gradient problem, which causes them hard to learn in the training process.

Although the above-mentioned machine learning algorithms are utilized for different applications and still have great performance, recent years have witnessed more complicated and adaptable approaches for different purposes of imputation in many applications. GAN-based algorithms are one of those approaches that have attracted the researchers' interest. Our exploration shows that there is an increasing research interest in GAN-based imputation algorithms. Figure 1 shows the number of published papers that contain "Generative adversarial networks" and "missing data imputation" as keywords in the Scopus database.

III. GAN-BASED DATA IMPUTATION METHODS

In this section, the main structure of the generative adversarial networks (GANs) is introduced. GANs are designed to produce synthetic data similar to real data. We then discuss the imputation methods derived from the main structure of GAN. This branch of deep learning has gained great attention in recent years because it not only imputes the missing values but also can generate artificial samples that are not the same as real ones. The generated artificial samples can be used as additional inputs for the supervised machine learning algorithms to increase their precision [89].

GAN has been introduced by Goodfellow et al. in 2014 [14] and used for a wide variety of applications including, but not limited to, image processing [90], fingerprint localization [91], speech recognition [92], video

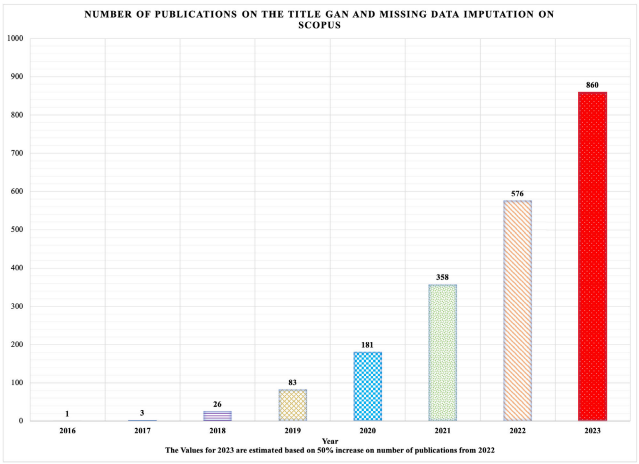


FIGURE 1. The number of papers in the Scopus database that have the keywords "Generative adversarial networks" and "missing data imputation" as keywords. The chart for 2023 is anticipated to be 50% higher than in 2022.

prediction [93], and especially data imputation [94]. GAN generally consists of two components, namely Generator (G) and Discriminator (D). Here, we describe both the training and generation procedure of the conventional GAN, shown in Figure 2. In the training phase, the noise always spikes the G component to convert it to real data. In other words, the noise and real data are the input and output of the G component in the training phase. The dominant goal of the G component is to alter the noise to the real data. Afterward, the D component learns to discriminate the real and fake data generated data. The training process of these two components is based on the following cost function presented in Eq. (1).

$$\min_G \max_D \; L(D,G) = E_{x\sim p_r(x)}\left[\log D(\mathbf{x})\right] + E_{z\sim p_z(z)}\left[\log(1-D(G(\mathbf{z}))\right] \quad (1)$$

Equation (1) can be seen as two dominant parts, to assign the true labels to the real and fake data. The G and D components are two functions, and they can be denoted by a multi-layer perceptron (MLP). The G component learns to alter the noise $\mathbf{z} \sim p_z(z)$ to the real data. It can be represented by $G(\mathbf{z};\theta_g)$ as a function, whose inputs are noise and outputs are generated fake data, in which θ_g indicates the parameters of the G component. The D component learns to discern real and fake data. This component can be represented by $D(\mathbf{x};\theta_d)$, whose inputs are real and fake data and outputs are class labels, in which θ_d indicates the parameters of the D component. In other words, this component has a binary classification architecture, in which zero and one show the fake and real data, respectively. These two components can be represented by two separate cost functions. The cost function of the D component, named $L(\theta_d)$ is presented in Eq. (2).

$$L(\theta_d) = E_{x\sim p_r(x)}\left[\log D(\mathbf{x};\theta_d)\right] + E_{z\sim p_z(z)}\left[\log(1-D(G(\mathbf{z};\theta_g))\right] \quad (2)$$

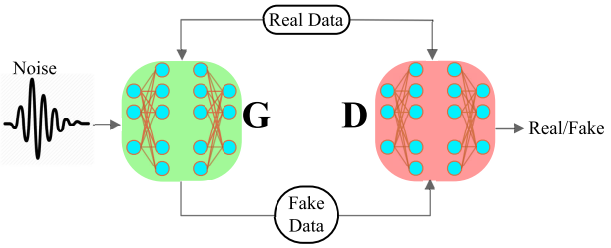


FIGURE 2. Structure of basic GAN proposed by Goodfellow et al.

Algorithm 1 (Training of G and D in GAN)

Require: number of epochs e; Number of Iteration of inner loop n; updating rates (α_g and α_d);

for $(i = 0; i < e; i{+}{+})$ **do**
  for $(j = 0; j < n; j{+}{+})$ **do**
    batch samples from noise $Z \in R^{B\times L} \sim p_z(z)$;
    batch samples from noise $X \in R^{B\times M}$;
    $L(\theta_d) = \frac{1}{B}\Sigma_{b=1}^{B}\log(D(X_b,\theta_d)) + \log(1 - D(G(2Z_b),\theta_g))$;
    $\xi_d = \frac{\delta}{\delta\theta_d}L(\theta_d)$;
    $\theta_d^{j+1} = \theta_d^t + \alpha_d\xi_d$
  end for
  batch samples from noise $Z \in R^{B\times L} \sim p_z(z)$;
  $L(\theta_d) = \frac{1}{B}\Sigma_{b=1}^{B}\log(1 - D(G(Z_b,\theta_g)))$
  $\xi_d = \frac{\delta}{\delta\theta_d}L(\theta_d)$;
  $\theta_g^{i+1} = \theta_g^t - \alpha_g\xi_g$;
end for

Equation (3) shows the cost function of the G component, named $L(\theta_g)$ as follows:

$$L(\theta_g) = E_{z\sim p_z(z)}\left[\log(1 - D(G(\mathbf{z};\theta_g))\right] \quad (3)$$

The parameters of the G and D components, θ_d and θ_g, are updated based on Algorithm 1 [14]. This algorithm is iterated till reaching a convergence. When the real and fake data cannot be recognized by the D net, we can conclude that convergence has occurred. Mathematically, the convergence occurs when $D(\mathbf{x};\theta_d) = 0.5$. As mentioned earlier, the G component can be used for generating artificial samples after the training phase. Another usage of GAN is to recognize artificial attacks or abnormal detection by the D component. However, we do not further discuss the applications of this component, since it is far from the scope of this paper. In summary, the G component after the training phase is ready to be utilized for producing the artificial samples with the spike noise $\mathbf{z} \sim p_z(z)$.

There are two main loops for updating the parameters of the G and D components in Algorithm 1. First, batch samples from the noise and samples of real data are presented to the inner loop for updating the parameters of the D component. The cost function of the D component is then calculated by the given samples. Then, the D component's parameters are updated based on the initiated rate. The same procedure is also conducted for updating the parameters of

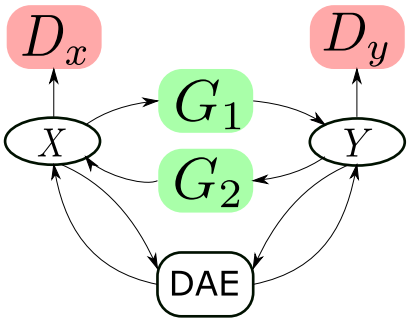


FIGURE 3. The architecture of VIGAN used for missing data imputation.

the G component. It should be noted that the parameters of the D component are updated e times compared with the G component's parameters.

GAN introduces a high opportunity for future research; however, it suffers from some issues including non-stability and gradient vanishing problems, and it can only generate unlabeled data. Therefore, other models are proposed to solve the mentioned issues. For instance, conditional GAN [16] and Wasserstein GAN [17] have been introduced to produce the data with class labels and to solve the instability problems, respectively. Since the goal of this paper is not to introduce the GAN models for data generation, the interested reader is referenced to [16], [17].

Although the GAN has been introduced to produce artificial samples, it can be used to impute the missing values by changing its evolved structures. In the following, we explain the other GAN-based models, which have been designed to deal with missing values in incomplete datasets. Since the main model of GAN had been designed in 2014, the derived techniques for imputation of missing values are very novel, and they belong to 2017 and after. Here, we discuss some of the GANs and their challenges for the imputation of missing values.

A. VIGAN

A GAN-based data imputation algorithm called view imputation with GAN (VIGAN) is proposed in [95]. This approach consists of two main parts including, a CycleGAN [96] and a denoising autoencoder (DAE), as depicted in Figure 3. The CycleGAN is used to convert samples with missing values to the target domain and vice versa, and the DAE is used to denoise the GAN outputs. These two networks (CycleGAN and DAE) are jointly trained till reaching a convergence. The CycleGAN has two generators ($G1$ and $G2$) and two discriminators (Dx and Dy). The x and y domains are the missing and non-missing parts, respectively. In CycleGAN, these two domains are converted to each other. The DAE receives the data from $G1$ and $G2$ and reconstructs them in an unsupervised manner, and then returns the reconstructed samples to these domains. When the training phase is completed, the $G2$ can impute the missing values for the given vectors.

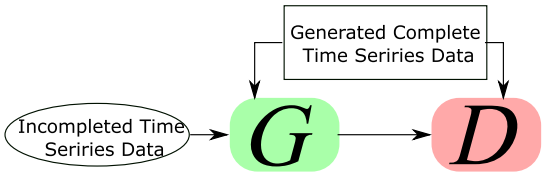


FIGURE 4. E2GAN structure used for missing data imputation.

B. I2-GAN

An adaptation of the conditional GAN to impute missing cardiac management resonance (CMR) slices is presented in [97], which is useful for image datasets. The method presented in [97] has been named image imputation-based GAN (I2-GAN) and is able to impute the missing slices conditioned on the information of vicinity slices. Initially, a regression network is utilized to predict the location of the missing slices and elicit the features of the adjacent slices. Then, the G component takes the pre-computed inner slice attributes as its conditioned input. Finally, the D component discerns the generated fake samples from the real images, and at the same time, authenticates the attributes of the inferred slices with that of real slices.

C. E2GAN

The missing values also occur in multivariate time series data. An end-to-end generative model (E2GAN) for the imputation of multivariate time series data is presented in [98]. The E2GAN imputes the incomplete time series using the adjacent generated complete time series. In the G component, E2GAN adds a random vector to the main sample and tries to convert it into a compressed vector with the help of an autoencoder, and then reconstructs it from the compressed vector to a complete time series. Their proposed method automatically learns the intrinsic features of the time series and attempts to reconstruct the temporal data. Finally, E2GAN uses the reconstructed sample to impute the missing values, as depicted in Figure 4.

D. GAIN AND VGAIN

A novel GAN-based data-imputing method called generative adversarial imputation nets (GAIN) is introduced in [99]. In their proposed method, the generator component G takes a real data vector, imputes the missing values conditioned on the really observed data, and gives in output a completed vector. Then, the discriminator component D gets a completed vector and tries to determine which element is really observed and which one is imputed. To learn the desired distribution in the G component, some additional information is deployed for the discriminator D in the form of a hint vector. The hint vector shows the pieces of information about the missingness of the real data to the D component, and D concentrates its heed on the quality of imputation for particular missing values. In other words, the hint vector assures the G component to be learned for generating data based on the actual data distribution. The architecture of their

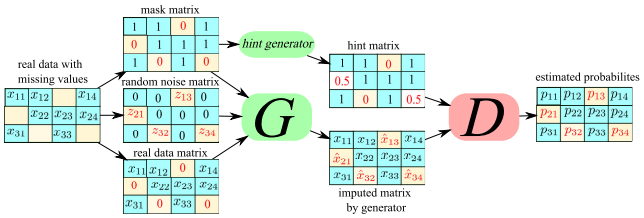


FIGURE 5. Structure of GAIN used for missing data imputation.

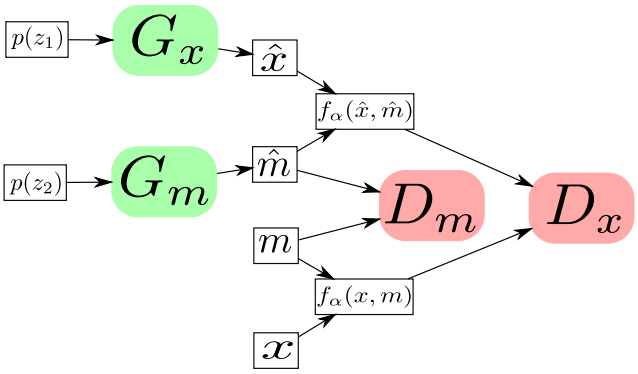


FIGURE 6. Structure of MisGAN used for missing data imputation.

proposed method is depicted in Figure 5. When the training phase is completed, the G component is able to impute the dataset. A new algorithm called VGAIN is introduced based on the combination of GAIN and variational autoencoder (VAE) [23]. In the generator, the encoder of VAE adds Gaussian noise to the data distribution for robust feature learning. Another variant of GAIN is called deconvolution GAIN (DEGAIN) [100] that adds a processing layer to remove the convolution for faster training of the generator.

E. MisGAN

A GAN-based framework for learning from an incomplete dataset, called MisGAN is proposed in [101]. The MisGAN trains with a complete data generator as well as a mask generator that models the missing values' patterns. An adversarially trained imputer is added to impute the missing values. The mask data is the same as the mask matrix, which is depicted in Figure 6 for the GAIN model; however, the mask matrices here are generated by a separate generator in a missing at-random process. Two pairs of generators and discriminators ((G_m, D_m) and (G_x,D_x)) are used for the mask and real data, respectively. The mask operator function f_α fills the missing values by a constant value α. Note that two generators have completely different noise distributions on their inputs p_{z1} and p_{z2}. The architecture of MisGAN is depicted in Figure 6.

F. HexaGAN

There are three major problems for classification tasks in real-world datasets:

- a) missing values

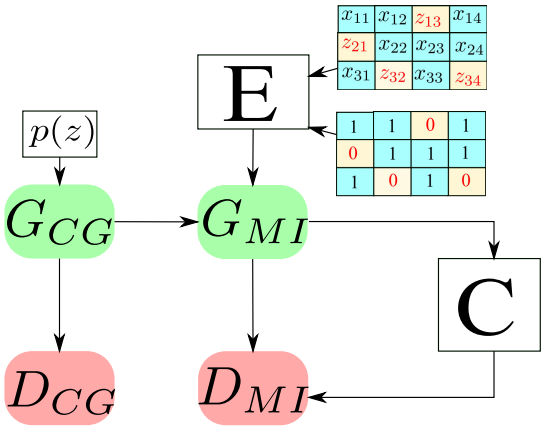


FIGURE 7. Structure of HexaGAN used for missing data imputation.

- b) missing labels (class labels)
- c) class imbalance

The class imbalance occurs when some classes have fewer samples compared with other classes. These three problems have been solved by several studies separately. Authors in [102] proposed a GAN-based data imputation algorithm, called HexaGAN, to cope with all three mentioned problems. They observe the mentioned problems from a single view to handle them jointly. HexaGAN framework has six components, which are linked together, as shown in Figure 7. The six components E, C, G_{MI}, D_{MI}, G_{CG}, D_{CG} have different duties, where the encoder E transfers the labeled and unlabeled data into hidden space, the classifier C estimates the class labels, the generator G_{MI} imputes the missing data, the discriminator D_{MI} discerns the missing and non-missing values and also class labels, the generator G_{CG} is used to construct the conditionally hidden vectors, and discriminator D_{CG} is used to determine whether the input data comes from original data or from G_{CG}.

G. TSDIGAN

Researchers in [103] introduced a GAN-based imputation framework for traffic sensor data (called TSDIGAN) to impute the missing data by generating synthetic samples. Although GANs have shown great achievement in generating synthetic images, producing traffic data using a GAN-based algorithm is a challenging issue. Because traffic data are time series, GAN-based algorithms should be justified for these kinds of tasks. TSDIGAN uses an encoding component that helps to extract the efficient temporal dependency of the time-series data, such that the time-series imputation problem is converted to an image imputation problem. The structure of TSDIGAN is shown in Figure 8.

H. IGANI

A GAN-based imputation algorithm for traffic time-series data called iterative GAN for imputation (IGANI) is proposed in [104]. According to the authors' claim, IGANI

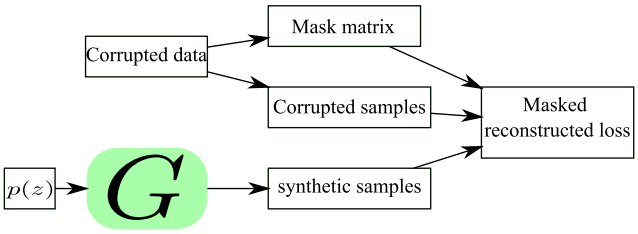


FIGURE 8. Structure of TSDIGAN used for missing data imputation.

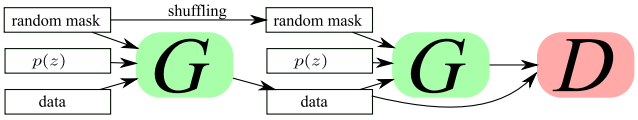


FIGURE 9. Structure of IGANI used for missing data imputation.

outperforms two of the previous GAN-based algorithms (i.e., GAIN and MisGAN) in terms of accuracy, stability, and simplicity of the model. IGANI method is simple and benefits a single pair of G and D components. The G component is iterated over imputed data, and the D component tries to identify the real and imputed data, respectively. Figure 9 shows the IGANI architecture.

I. BiGAN

Multivariate time series are utilized in many classification and regression problems, and recurrent algorithms are extensively employed for these problems. The common recurrent algorithms convert time-series data elements to equal-length vectors. Nevertheless, time-series data can have different lengths and missing values in practice, which diminishes the performance of the algorithms. Authors in [105] proposed a GAN-based data imputation algorithm, called bidirectional GAN (BiGAN). Bi-GAN combines the bidirectional recurrent network with GAN. In fact, the generator component plays the bidirectional recurrent network role that receives real data with missing values and imputes them. On the other side, the discriminator tries to distinguish the real and imputed data received from the G component. The BiGAN has two main advantages compared with other methods in the field of imputation:

- (a) a single model is utilized for both imputation and prediction
- (b) it can impute the time series data with different lengths

The structure of BiGAN is depicted in Figure 10. In addition to the real data with missing values, two additional data are given to the G component, namely forward time decay and backward decay. These two data are created based on the location of missing and non-missing values. In order to create the forward time decay, the numbers start with zero at the beginning of time and then increase consecutively till reaching a missing value or two non-missing values. In these two situations, it will restart again. In the backward, the same process is done inversely.

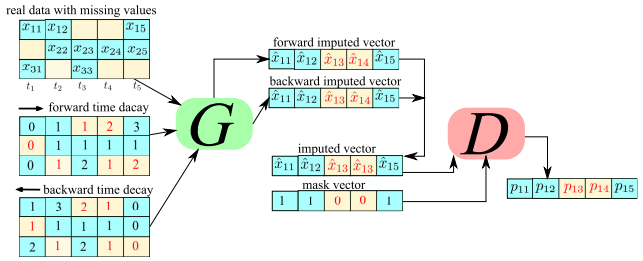


FIGURE 10. Structure of BiGAN used for missing data imputation.

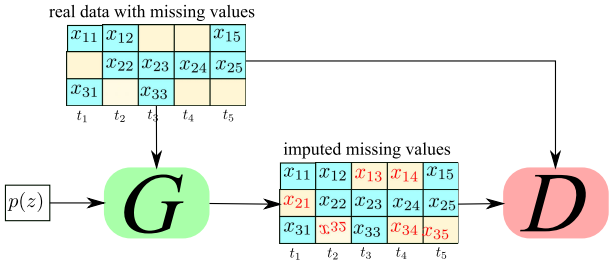


FIGURE 11. Structure of MTSIGAN used for missing data imputation.

J. MTSIGAN

Authors in [106] designed a novel GAN-based algorithm to deal with missing values in incomplete time series data, called multivariate time series with GAN (MTSIGAN). As mentioned before, some values in time series data are sometimes missed due to their nature during the collection phase. They designed a modified gate recurrent unit, to model the inconsistency of time series in incomplete datasets. After the training phase of the GAN with the modified gated recurrent unit (GRU) cell, the noise in the input of the G component spikes it to impute the missing values. In fact, MTSIGAN benefits from the main structure of GAN; however, the layers are modified with the GRU. The following figure indicates the structure of MTSIGAN.

K. ARGAN

Conditional GAN-based data imputation algorithm for time series data, called adversarial recurrent time series (ARGAN) is proposed in [107], in which the G and D components are conditioned on the non-missing values besides the labels. ARGAN uses a justified bidirectional recurrent neural network architecture for the G component, which is used to impute the missing values by exploiting the temporal and non-temporal information from the time series data. The D component is employed to discern the imputed and complete data samples, which can help the G component generate more reliable samples. Figure 12 shows the structure of the ARGAN, where M is the mask matrix that shows the location of missing and non-missing values, and $B \in \{0, 1\}$ is a random value that has Bernoulli distribution, which is used to construct the hint matrix. Another conditional GAN-based imputation called *pix2pix* is presented in [108], in which the

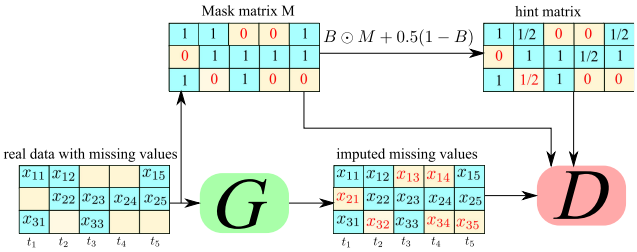


FIGURE 12. Structure of ARGAN used for missing data imputation.

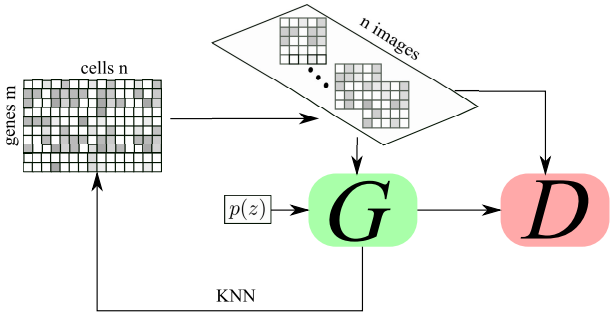


FIGURE 13. Structure of sclGAN used for missing data imputation.

proposed network learns time series features and exploits spatial and temporal features for imputation.

L. scIGAN

A GAN-based algorithm for imputation of single-cell RNA-sequencing (scRNA-seq), called scIGAN, which utilizes the synthetically generated cells instead of observed ones is proposed in [109]. Some false zero values disrupt the recognition accuracy of the homogeneous cell population in the scRNA-seq data, known as dropouts. In the first step, the real single-cell data are converted to reshaped images, and then they are presented to the input of G and D components. The G component produces synthetic images using latent noise. As usual, the D components assess whether the images are synthetic or real. After the training phase, the G component is utilized to generate image-based scRNA-seq data for any type. Then, the KNN method is used for the imputation of the missing values (dropouts) for the same cell type in the real data. According to the designers' aim, the sclGAN avoids overfitting for cell types in a large population. Figure 13 shows the structure of the scIGAN. The STA-GAN [110] is proposed for satellite data imputation and uses the spatiotemporal attention (STA) for training the G. In STA-GAN also the correlation in images is used for missing data imputation.

M. TTI-GAN AND CM-GAN

Authors in [109] proposed a GAN-based imputation algorithm for missing traveling times data, called TTI-GAN, which is used for traffic management. TTI-GAN can generate the traveling times between links with a few samples using the spatiotemporal correlations and modeling of travel time

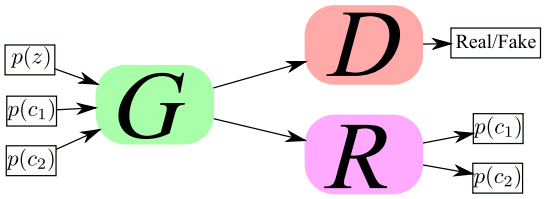


FIGURE 14. Structure of TTI-GAN used for missing data imputation.

distribution. TTI-GAN benefits from a simple structure similar to GAN, and the difference is that it has a recognition component R beside the G and D components, which helps to recognize the statistical features of time series data. Also, besides the latent noise, two additional vectors, c_1 and c_2, are used as hidden codes, and the R component tries to reconstruct those distributions. According to the designers, to reach good performance, z, c_1, c_2 should have gamma, normal, and uniform distributions, respectively. Figure 14 shows the structure of TTI-GAN for the imputation of traveling times in traffic data. Another similar GAN-based traffic management is presented in [111]. In CM-GAN the authors consider complete data missing (CDM) and combine the cross-modal data fusion with the GAN technique to construct a cross-modal data generator [112]. They use the spatiotemporal correlation of time series for industrial systems.

N. 3D-cGAN

Neuro images such as magnetic resonance imaging (MRI) and positron emission tomography (PET) are used in a wide variety of applications for brain disease diagnosis. For instance, they can be used for Alzheimer's disease (AD) diagnosis. However, similar to many datasets, it suffers from missing values. One simple way to cope with this issue is to remove the subjects with missing values, which leads to decreasing the number of training data. Since MRI and PET are obtained from the same subject, they have relevant information. According to this fact, in [113] the authors propose a GAN-based data imputation algorithm, called 3D cycle-consistent GAN (3D-cGAN) for the imputation of the missing values. In summary, 3D-cGAN imputes the missing values of the PET data using the corresponding relevant MRI data. The structure of 3D-cGAN is depicted in Figure 15, which consists of two pairs of G and D components. Another 3-D GAN framework called *MDA-GAN* is presented in [114] in which the authors use the spatial continuity of the data and tanh cross entropy (TCE) loss to reveal the missing values.

Summary of the reviewed GANs for missing data imputation is presented in Table 2.

O. CATEGORIZATION

We identify four major groups among the GAN-based data imputation algorithms including, time-series, regular, specific, and image-based GANs as depicted in Figure 16. The *regular* GAN-based imputation methods handle the

TABLE 2. The summary of the reviewed GAN-based missing data imputation methods.

Paper	Year	Method	Application	Description	Challenge
Shang et al. [95]	2017	VIGAN	General (Regular)	Combined of two networks (CycleGAN and DAE)	Uses dynamic aspect-aware to purify the imputed values by decreasing the noise effects in highly noisy datasets. However, it may not be appropriate for multi-view clustering, since it does not consider the difference between random and non-random effects in missing data
Xia et al. [97]	2021	I2-GAN	Images	An adaptation of the conditional GAN to impute missing cardiac management resonance slices	I2-GAN uses additional information on its input for providing a better imputation result. However, it cannot be applied to many variants since most of the time the conditioned information is not available
Luo et al. [98]	2019	E2GAN	Time Series	E2GAN adds a random vector with the help of an autoencoder	Finds time-series dependency, however, it is hard for an RNN to capture correlations of distance variables and different time-series information especially in small datasets
Yoon et al. [99]	2018	GAIN	General (Regular)	The hint vector assures that the generator is trained (learned) based on the actual data distribution.	Uses a hint vector for focusing the attention of the discriminator component on the imputation quality. Nevertheless, it suffers from the instability of training for block missingness
Miao et al. [23]	2023	VGAIN	General (Regular)	Combines GAIN and variational autoencoder	It might suffer from the instability of training for block missingness
Shahbazian et al. [100]	2022	DeGAIN	General (Regular)	Adds a deconvolutional layer to GAIN to improve the training	Suffers from the instability of training for block missingness
Gupta et al. [101]	2019	MisGAN	Images	Uses two generators and discriminators that have completely different noise distributions on their inputs	Although is able to handle high-dimensional incomplete data, the training is an issue
Hwang et al. [102]	2019	Hexagan	General (Regular)	Considers three problems of missing data, class imbalance, and missing labels to solve jointly	It can only be utilized for classification problems in which the class labels have been missed
Huang et al. [103]	2020	TSDIGAN	Time Series Images	Converts temporal dependency of the time-series data to images	It needs a high number of samples, since it converts the time-series data to images
Kazemi et al. [104]	2020	IGANI	Time Series	Iterative GAN that uses two generators and has high-performance in time-series	It supports only one type of missing data and does not consider the correlation of missing data
Gupta et al. [105]	2020	BiGAN	Time Series	Supports different lengths of time-series data and has joint training for both imputation and prediction	It is hard to extract the information since it uses a global RNN to capture the information. Besides, the RNN might not be able to capture the correlations of distance variables
Luo et al. [106]	2018	MTSIGAN	Time Series	Captures inner-class correlation, distribution, and temporal correlations	Follows the auto-regressive imputation, which causes error propagation during the training phase and damages the performance of imputation for long sequences
Yang et al. [107]	2020	ARGAN	Time Series	Uses a justified bidirectional RNN architecture exploiting the temporal and non-temporal information from the time series data	It uses the labels (metadata) while we don't have access to the labels of conditional information
Almeida et al. [108]	2023	pix2pix GAN	Time Series Images	Convert time series data to images for missing	Training complexity is an issue
Zhang et al. [109]	2021	scIGAN	Specific Images	Utilizes the synthetically generated cells instead of observed ones. Also uses KNN to prevent overfitting in large populations	Good performance in high-noisy datasets, however, it is more appropriate for small datasets
Wang et al. [110]	2023	STA-GAN	Specific Images	Uses spatiotemporal Attention mechanism based on the graph attention network to learn features for capturing both short-term temporal and dynamic spatial dependence	The training process is hard
Zhang et al. [109]	2021	TTI-GAN	Time Series	Uses the spatiotemporal correlation for time series in traffic data	Low performance with a high missing rate
Kang et al. [112]	2023	CM-GAN	Time Series	Combines the cross-modal data fusion technique and the deep adversarial generation technique	requires large datasets to train the model
Pan et al. [113]	2018	3D-cGAN	Images	Uses two generators and two discriminators	Does not take into account the spatial consistency between each pair of PET and MRI obtained from the same subject
Dou et al. [114]	2022	MDA-GAN	Images	use the spatial continuity of the data and tanh cross entropy (TCE) loss	The computational complexity is high for training

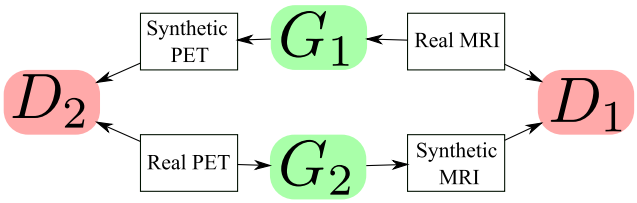


FIGURE 15. Structure of 3D-cGAN used for missing data imputation.

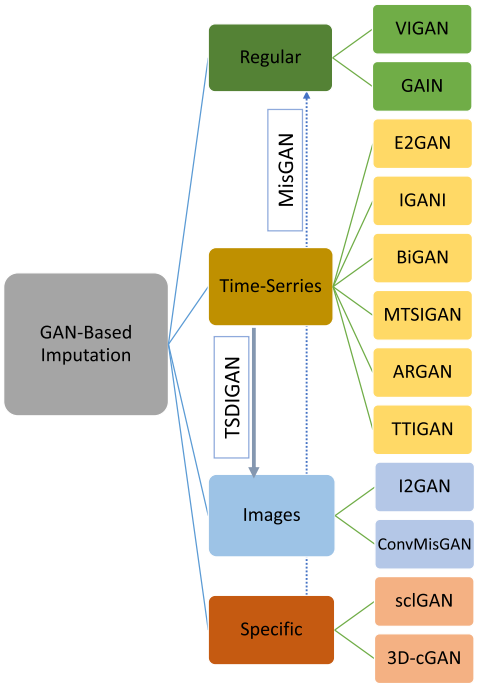


FIGURE 16. Major groups among the GAN-based algorithms including, time-series, regular, specific, and image-based GANs.

typical datasets where there is no additional information such as spatiotemporal dependencies. VIGAN and GAIN fall into this category. The specific GAN-based imputation algorithms deal with particular problems. For instance, 3D-cGAN has been developed for the imputation of PET data according to the MRI data, which are drawn from the same subject and have relevant information. This particular type of imputation can rarely be utilized for other fields of studies, and we marked them as *specific* in our categorization.

GAN-based time-series imputation algorithms cope with those datasets that have temporal dependencies. These algorithms regularly use a time-based neural network such as RNN, LSTM, and GRU on their components. The challenge here is how to use these time-based networks to extract sequential information. The time-based neural networks are regularly utilized in the G component for imputing the missing values after training. Image-based GANs regularly use spatial information of images to impute the missing pixels. ConvMisGAN is a great example of this categorization in which convolutional filters are used in the components to extract high-level features of images.

Some GAN-based algorithms cannot be categorized into one type of these four groups. For example, MisGAN first converts the images to vectors and fully connected neural networks, which handle the regular imputation problems, are utilized for imputation purposes. As another example, TSDIGAN first converts the sequential data to images, and an image-based GAN is used to deal with the imputation problem. By doing so, the temporal information is converted to spatial information.

IV. EVALUATIONS

Different strategies are used to evaluate imputation techniques. Here, we present some of the common metrics and methods that are used for the assessment of data imputation algorithms.

A. EVALUATION METRICS

Controlled missingness and calculating the difference between the original values in the collected dataset and the estimated values in the simulated incomplete dataset is one of the simplest yet most effective approaches for evaluating the efficacy of imputation algorithms. It should be noted that in practice we usually do not have access to an entire dataset to model the missingness process. There are two types of attribute values, namely, discrete and continuous. In Table 3, we have summarized some of those metrics related to discrete or continuous values.

- **Root-mean-square error (RMSE)**: RMSE is used for continuous values and measures the average squared differences between real values and imputed values for an incomplete dataset. When the size of the error is important, RMSE is helpful because it penalizes bigger errors more severely. Outliers can have an impact on the RMSE if they are present in the data.
- **The percentage of correct predictions (PCP)**: PCP is usually utilized to assess the imputation results for the discrete datasets. In fact, PCP assesses the precision of categorization models. PCP does not take into account the magnitude of the changes between imputed and true values, which is significant when evaluating the accuracy of imputation methods for continuous variables.
- **Structural similarity index measurement (SSIM)**: Generally, this metric is used to measure the quality of images. In the context of data imputation, the SSIM is used to measure the quality of imputed pixels. Assuming that x and y are two images, one of them has perfect quality, the goal is to assess the quality of the second image with the SSIM. Higher SSIM values indicate better imputation quality.
- **Peak signal-to-noise ratio (PSNR)**: The PSNR calculates the maximum possible power of the signal to the noise between two images x and y to assess the quality of the imputed images compared with real images. In fact, it measures the amount of noise or distortion in the reconstructed data as compared to the original reference

TABLE 3. Evaluation metrics of discrete or continuous values used to assess data imputation algorithms' performance.

Metric	Formula	Description
RMSE [103]	$RMSE = \sqrt{\frac{1}{\hat{N}} \sum_{i=1}^{\hat{N}} (y_i - \hat{y}_i)^2}$	$\hat{N}$is the number of missing values, $y_i$is the real missing value, and $\hat{y}_i$ is the imputed value.
PCP [103]	$PCP = \frac{\text{number of true predictions}}{\text{numberofallsamples}} \times 100$	-
Structural similarity index measurement (SSIM) [97], [115]	$SSIM(x,y) = \frac{(2\mu_x\mu_y+C_1)(2\sigma_{xy}+C_2)}{(\mu_x^2+\mu_y^2+C_1)(\sigma_x^2+\sigma_y^2+C_2)}$	C_1 and C_2 are constant values that should be determined, μ_x and μ_y indicate mean of intensity in a local image patch, σ_x^2 and σ_y^2 denote the variance of patches, andσ_{xy} represents the correlation coefficient between corresponding patches between two images x and y
Peak signal-to-noise ratio (PSNR) [97], [116]	$PSNR = 10\log_{10}\frac{(2^n-1)^2}{RMSE}$	n is the number of to represent pixels' values (for 8 bit images n equals to 8).
Frechet Inception distance (FID) [101], [117]	$d^2((m_1,C_1)(m_2,C_2)) = \|m_1-m_2\|_2^2 + \text{Tr}(C_1+C_2-2(C_1C_2)^{1/2})$	m_1 and m_2 denote the mean of the real and imputed data, respectively. C_1 and C_2 indicate the covariance metrics of real and imputed data respectively
F1-score [102], [118]	$F_1\text{ score} = \frac{2\cdot TP}{2\cdot TP+FP+FN} = 2\frac{\text{precision}\cdot\text{recall}}{\text{precision}+\text{recall}}$	TP, FP and FN indicate the True Positive, the False Positive and the False Negative, respectively.
Mean absolute error (MAE) [103], [119]	$MAE = \frac{1}{\hat{N}} \sum_{i=1}^{\hat{N}} \|y_i - \hat{y}_i\|$	$\hat{N}$ is the number of missing values, y_i is the real missing value, and $\hat{y}_i$ is the imputed value.
Area under ROC curve or AUC score [106], [120]	$AUC = \frac{\sum_{i=1}^{m}\sum_{j=1}^{n} 1_{x_i>y_j}}{mn}$	x_i and y_i represent the positive and negative values of a binary classifier.

data. The larger value of the PSNR shows a better performance for the imputed pixels.

- **Frechet Inception distance (FID)**: The FID converts a group of imputed samples to a feature space using a particular inception net layer. Assuming the converted layer as continuous multivariate Gaussian distribution, the mean and covariance are predicted for both the imputed and real samples. A lower FID number suggests that the generated images are of higher quality and diversity. FID considers both the visual quality of generated images and their distributional closeness to real-world images.
- **F1-score**: For binary classification, the F1-score is used to assess the test accuracy, which is calculated as the harmonic mean of precision and recall. The precision is calculated by dividing the number of true predicted positive values by the total number of samples, and the recall is calculated by dividing the number of true predicted positive values by the number of those samples that should be predicted as positive. The F1-score is especially beneficial when the dataset is unbalanced, which means that the number of cases in different classes varies dramatically. F1-score is normally employed in classification tasks with available class labels and may not be directly applicable to evaluating missing data imputation techniques unless the imputation job involves predicting categorical variables or class labels.
- **Mean absolute error (MAE)**: This metric is similar to the RMSE and measures the error between the true and predicted values. It is unaffected by error direction and considers all errors identically, whether positive or negative in form. A lower MAE score denotes more accuracy because it shows a smaller average gap between predicted true values. MAE can be used to predict continuous numerical values. However, the RMSE is a better choice compared with the MAE when the error has a normal distribution [119]. The MAE is less sensitive to outliers compared with RMSE.
- **Area under ROC curve (AUC)**: The receiver operating characteristic curve (ROC) is a plot, which shows the prediction ability of a binary classifier. The ROC curve is plotted based on two predefined variables, called true positive rate (TPR) and false-positive rate (FPR). Alternatives for TPR used in the literature are recall and detection probability, whereas for FPR it is the probability of false alarms. The AUC metric is used to calculate the area under the ROC curve, which is used for assessing the binary classifiers' performance. It should be noted that AUC is only applicable to binary classification tasks and may not be suitable for evaluating models in regression or multiclass classification scenarios. The AUC value ranges from 0 to 1, where, AUC = 1 denotes a flawless classifier capable of distinguishing between positive and negative cases

and AUC = 0.5 denotes that the classifier performs no better than random chance. AUC greater than one suggests better classification ability, whereas values closer to one indicate better discrimination between the two groups. When the dataset is imbalanced, the AUC measure is very beneficial because it is less affected by class distribution. It gives an aggregated measure of the model's performance over many thresholds, allowing for a thorough evaluation of its discriminatory strength.

There are other metrics such as ARMSE [24] that combine the RMSE and accuracy error (AR). AR is a metric that is usually used for categorical features in the datasets [121]. Smaller values of ARMSE mean that the algorithm performs better at missing data imputation. It should be mentioned that not all these metrics are used together. For instance, the authors in [99] use the RMSE and AUC for their evaluations, and the authors in [100] use the FID and RMSE metrics. While the used metric in [105] is MAE.

B. DATASETS

An important part of the evaluation methods besides the metrics is considering the proper scenarios and datasets. In [121] the authors use the *breast*, *spam*, *letter*, and *news* datasets while in [100] only *letter* and *spam* datasets are used. Authors in [105] use the *Air quality*, *MIMIC*, and *EHR* datasets for performance evaluations. In image recovery applications, the datasets are defined based on specific needs. For instance, in [95] the authors use *CMR images from the UK Biobank (UKBB)* dataset. In some studies, the authors do not use the existing dataset and focus more on the self-generated datasets. For example, in [122] the authors generate a dataset by discretization to enable the specific missing scenarios. As shown in Table 4, we summarize some of the common datasets used in the GAN-based data imputation models. Some of these datasets have missing values, and therefore, we cannot directly use them to evaluate the performance of imputation algorithms. On the other side, some of them are complete and may be used to simulate the missingness of the data and to calculate the performance of imputation models. In Table 4, we mention the number of samples and the dataset description beside their access link. An important feature in evaluations is related to the data type that we address in Section V.

C. EXPERIMENT RESULTS

In the following, we use the breast, spam, and letter datasets to compare a regular GAN-based imputation algorithm[2] (GAIN [99]) with other machine learning imputation algorithms such as AE [123], EM [124], MisForest [24], and MICE [125]. We assess these algorithms with 5-fold cross-validations in terms of RMSE, which directly measures the error distance, and FID, which takes the distribution of imputed values into account. We use the breast, letter, and spam datasets for comparing algorithms, and samples are missed at the rate of 20%. Table 5 shows the performance of algorithms in terms of RMSE. It should be highlighted that comparing GANs in different categories is not fair. For example, GANs developed for missing data imputation in time series are projected to outperform regular GANs in specialized applications such as traffic management.

[2]We use GAIN [99] for comparison because it is a regular GAN-based algorithm and can be compared with others in most of the datasets and applications. However, it does not mean that GAIN is the best approach among the GAN-based imputation algorithms.

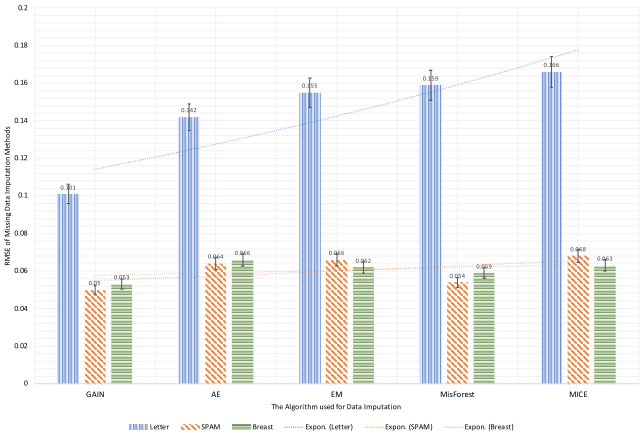


FIGURE 17. The illustration of the RMSE evaluation, is presented in Table 5. The Exponential Trend and 5% error bars are also presented.

As can be seen in Table 5 and Fig. 17, the GAN-based imputation algorithm has superior performance compared with others in all three datasets. Since the RMSE does not take the distribution of imputed values into account, it may be better to use other metrics to expose the ability of algorithms. Among the introduced metrics, some of them, such as SSIM and PSNR, are mostly used for image datasets, and the other ones, such as F1-score and AUC, are used to assess the classification accuracy after the imputation stage. Since we have access to the real values of missing data in the datasets, we do not use these precision metrics. The FID metric may be the best choice among them since it considers the distribution of imputed values. Table 6 and Fig. 18, show the comparison of algorithms in terms of FID. Table 6 shows that the GAN-based imputation algorithm has better performance compared with others in terms of FID.

V. DISCUSSION: OPPORTUNITIES AND CHALLENGES

This section will investigate how data type influences the performance of missing data imputation strategies. By presenting a useful benchmark tool, we also analyze the impact of missing data volume on algorithm performance. We discuss the challenges related to the GANs and make some remarks about computational complexity.

A. EFFECT OF DATA TYPE

The literature reports three types of missing data, called missing at random (MAR), missing completely at random (MCAR), and not missing at random (MNAR) [43]. MNAR occurs when the probability of happening a missing value

TABLE 4. The summary of datasets used in the GAN-based data imputation models.

Dataset Name	Dataset Description	Access Link
MNIST	The MNIST dataset consists of human handwritten digits, which is utilized for testing the wide variety of deep learning models. This dataset contains 60,000 training samples and 10,000 testing samples.	http://yann.lecun.com/exdb/mnist/
Predicting Mortality of ICU Patients (PhysioNet)	This dataset consists of 4000 ICU stays, each of which contains multivariate time series data with 42 attributes. The goal of this dataset is to predict the health state of patients from the time series records (binary classification). Since this dataset contains the missing values, it is impossible to assess the imputation methods directly. However, some metrics, such as AUC can be used to compare the algorithms.	https://physionet.org/content/challenge-2012/1.0.0/
KDD CUP 2018	The goal of this dataset is to forecast the competition, which was extracted from the KDD CUP challenge in 2018. There are 282 hourly collected time series data, labeled by city name, air quality, and station name. Also, 15 percent of this dataset includes the missing values.	https://zenodo.org/record/3893504#.X8_bodIzbIU
Breast	This dataset contains 569 samples and 30 input features. The goal is to predict the type of cancer (malignant and benign).	https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic)
Spam	This dataset consists of 4601 samples and 57 input features with the goal of predicting the spam email based on input features.	http://archive.ics.uci.edu/ml/datasets/Spambase/
Letter	This dataset consists of 20000 samples and 16 input features with the goal of predicting 26 English capital letters alphabet based on pixels. Usually, the first 16000 samples are used as training and the rest as testing.	https://archive.ics.uci.edu/ml/datasets/Letter+Recognition
Credit	This dataset consists of 30000 samples and 23 input attributes with the goal of predicting credible or un-credible clients based on input attributes.	https://archive.ics.uci.edu/ml/datasets/default+of+credit+card+clients
News	This dataset consists of 39797 samples, and 58 input features and the objective is to forecast the number of shares in social networks based on the given input features.	https://archive.ics.uci.edu/ml/datasets/Online+News+Popularity
Urban traffic speed	This traffic dataset consists of 214 road segments that have been collected in Guangzhou for two months. It has a 3D array structure on 214*61*44, where the first dimension is the road segments, the second dimension is the day, and the third dimension is the time window. Therefore, the total number of observations equals to 1,855,589. Also, 1.29 percent of this dataset contains the missing values.	https://zenodo.org/record/1205229
MIMIC-III	This is an open-source large dataset consists of more than 40,000 patients. The patients stayed in critical care units for two years in the Beth Israel Deaconess Medical Center.	https://physionet.org/content/mimiciii/1.4/
Air quality	This dataset consists of 9358 samples and 15 input features, which have been averaged hourly from an air quality multi-sensor device. The data has been collected for one year in a very contaminated environment in one of the Italian cities. This dataset consists of missing values marked by -200.	https://archive.ics.uci.edu/ml/datasets/air+quality
ADNI (1 and 2)	This is an Alzheimer's disease neuroimaging initiative dataset, consists of four subject categories: 1- AD 2- HC 3-pMCI, and 4- sMCI. The ADNI-1 dataset consists of 821 subjects (AD, HC, pMCI, and sMCI have 199, 229, 167, and 226 subjects, respectively), and ADNI-2 consists of 636 subjects (AD, HC, pMCI, and sMCI have 159, 200, 38, and 239 subjects, respectively).	http://adni.loni.usc.edu/

TABLE 5. Comparing a GAN-based imputation algorithm (i.e., GAIN [99]) with other machine learning algorithms such as AE [123], EM [124], MisForest [24], and MICE [125] in terms of RMSE.

	GAIN	AE	EM	MisForest	MICE
letter	0.101	0.142	0.155	0.159	0.166
spam	0.050	0.064	0.066	0.054	0.068
breast	0.053	0.066	0.062	0.059	0.063

TABLE 6. Comparing a GAN-based imputation algorithm (i.e., GAIN [99]) with other machine learning algorithms such as AE [123], EM [124], MisForest [24], and MICE [125] in terms of FID.

	GAIN	AE	EM	MisForest	MICE
letter	0.421	0.678	0.781	0.798	0.820
spam	4268.252	4391.265	4432.540	4294.132	4511.178
breast	247.65	352.334	310.191	319.581	338.490

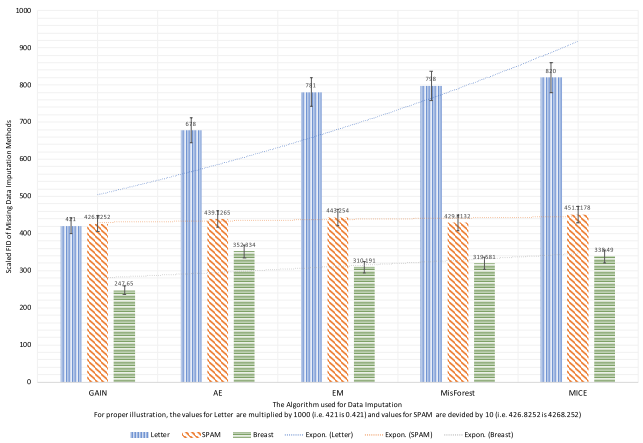


FIGURE 18. The illustration of the FID evaluation presented in Table 6. The Exponential Trend and 5% error bars are also presented.

depends on the attribute's value. The MCAR occurs when the probability of having the missing value depends on neither the known values nor missing data. This implies that the causes of the missing data are unrelated to the data. An example of MCAR is when a random sample of a population is taken, where each member has the same chance of being included in the sample. The (unobserved) data of members of the population that were not included in the sample are MCAR. If the missingness depends on the value of the sample but does not depend on the missing data, it is called MAR. An example of MAR is when a sample from a population is taken, where the probability to be included depends on some known property. MAR is more general and more realistic than MCAR.

Evaluation results of different ML-based imputation algorithms prove that the best results are usually gained working with the MCAR case [23], [122]. The performance of the algorithm under the MAR scenario is almost the same as MCAR for ML-based algorithms. However, experimental results in the literature show that the GAN-based algorithms outperform other existing methods in terms of accuracy [23], [122]. It is important to note that the MAR simulation is a difficult task. The problem arises during feature correlation, which requires determining the feature selection. The features chosen vary based on the dataset. All of the techniques perform almost poorly when compared to the MNAR missing data type, and while GANs outperform other existing algorithms, the result is still far from optimal. The reason for this is simple: in the instance of MNAR, the correlation is confined to the missing value. As a result, accurately anticipating the missing data is extremely challenging. One gap in the literature is a more thorough study of MAR circumstances rather than MCAR.

B. EFFECT OF MISSING DATA FRACTION

The impact of missing data fractions on performance is a significant problem that is not well addressed in existing studies. In this survey, for example, performance is evaluated with 20% missing data using the MCAR scenario. There has been very little research into the performance of ML-based data imputation algorithms in different data types and missing data fractions. The authors of [126] conduct various experiments while ignoring the inherent uncertainty of imputed data. Their assessments are based on F1 score and RMSE on 1%, 10%, 30%, and 50% missing data with MCAR, MNAR, and MAR. According to their findings, in the majority of studies, KNN outperforms other algorithms, although the GAIN technique performs badly. It's because, as the paper's authors point out, the hyperparameters aren't optimal for GAIN training [126].

Another experiment published by the authors in [105] compares the performance of Bi-GAN with KNN and Mean techniques for 10%, 20%, 30%, 40%, and 50% MAE. Their findings reveal that the GAN outperforms the KNN in all missing data fractions in time-series data. The results in [105] also confirm that performance drops slightly as the fraction of missing data increases. Experiments in [23] confirm the superiority of GANs (GAIN and VGAIN) in different missing data percentages over previous ML-based approaches. The discovery of [23] verifies the reported results of [105] that raising the missing data proportion reduces the performance of imputation algorithms, but not significantly. As a result, a benchmark for evaluating GAN-based missing data imputation algorithms remains an open challenge. Despite the fact that there are several practical difficulties to be solved. For example, comparing all GANs based on their various uses and designs seems infeasible and they should be compared in each category. Other difficulties include selecting the appropriate evaluation metrics and data type scenarios (MCAR, MNAR, MAR). Open-source tools are essential for objective evaluation. The "jenga" is one example that might be extended for GAN-based algorithms [127].

C. COMPUTATIONAL COMPLEXITY

The finding in the reviewed studies confirm the superiority of GAN-based data imputation. However, this performance is

highly dependent on hyperparameter optimization. In other words, the GANs are generally known as hard to train, and therefore some studies try to improve the training of GANs in terms of computational complexity [128]. For example, the authors in [129] propose a method named *Sim-GAN* that is applicable only in microarray array data for cancer analysis. In Sim-GAN three similarity metrics (structural, functional similarity, and semantic) are applied to a small subset of the dataset to reduce the computational complexity of the generator and avoid training on the entire dataset.

For non ML-based methods such as *Mean*, the computational complexity is in the order of $(d_m.n)$. For ML-based algorithms such as *MICE* the computational complexity is proven to be in the order of $(c.d_m.d^2.n^2)$, where c is the number of iterations, d_m is the number of incomplete features, d is the number of features and n is the number of samples [23]. Although the computational complexity analysis is not performed for GAN-based missing data imputation, it could be said that the GANs perform better compared to the earlier ML-based methods in the cost of more computational complexity. For instance, based on the findings on e wall-clock run time in [126], the time taken for the training on *Mean/mode*, *KNN* and *GAIN* are 0.005, 41, and 878 respectively. Although this gap is shorter in the inference phase and is reported as 0.029 for *Mean/mode*, 7 for *KNN*, and 137 for *GAIN*.

D. ALGORITHM-RELATED CHALLENGES

GAN-based data imputation algorithms outperform the existing ML-based algorithms, however, they also have disadvantages besides many advantages. This is why GAN-based algorithms are useful for some applications and inappropriate for others. For example, VIGAN uses dynamic aspect-aware (DAE) to purify the imputed values by decreasing the noise effects in highly noisy datasets. However, it may not be appropriate for multi-view clustering, since it does not consider the difference between random and non-random effects in missing data [130]. I2-GAN uses additional information on its input for providing a better imputation result. However, it cannot be applied to variant algorithms, since most of the time the conditioned information is not available. E2GAN can find time-series dependency, however, it is hard for an RNN to capture different time-series information in small datasets. Besides, the RNN might not be able to capture the correlations of distance variables [131].

GAIN uses a hint vector for focusing the attention of the D component on the imputation quality. Nevertheless, it suffers from instability of training for block missingness [101]. MisGAN could handle high-dimensional incomplete data. However, it is hard for MisGAN to be trained [132] and only focuses on multi-color convolution and attention-stacked in image datasets and may not be useful for other fields [133]. HexaGan can cope with three problems together: missing values, missing labels, and class imbalance. While it can only be utilized for classification problems assuming that the class labels have been missed. TSDIGAN uses the ability of images to extract high-quality data. However, it needs a high number of samples, since it converts the time-series data to images. IGANI has a high performance with a high missing rate. However, it assumes that the data have been missed at random. BiGAN deals with different lengths of time-series data and also has joint training for both imputation and prediction. However, it is hard for BiGAN to extract the information since it uses a global RNN to capture the information. Besides, the RNN might not be able to capture the correlations of distance variables [131]. MTSIGAN captures inner-class correlation, distribution, and temporal correlations. Nevertheless, it follows the auto-regressive imputation, which causes error propagation during the training phase and damages the performance of imputation for long sequences. ARGAN takes advantage of conditional GAN and RNN to exploit the information from the data as much as possible. For instance, when we collect data relating to the temperature, there is a hidden pattern between the time of collecting data and the time of recording the temperature in summer or winter. However, usually, we don't have access to the labels of conditional information [134]. sclGAN has a good performance in high-noisy datasets, however, it is more appropriate for small datasets. TTIGAN has a good performance without sufficient information, while it also has low performance with a high missing rate. 3D-cGAN has the desired performance in the studied problem. However, it does not take the spatial consistency between each pair of PET and MRI obtained from the same subject [135].

One of the reported challenges with the GAN-based algorithms is their instability in optimization which is addressed as the mode-collapse problem [136]. Mode-collapse in GANs means that the imputed data does not reflect the diversity of the underlying observed data distribution. To tackle this issue the authors in [23] propose to improve the robustness by introducing excess Gaussian noise as the input to the generator and combining the autoencoder with the existing GAIN method. However, the need for better solutions to this problem is still an open challenge.

To address the existing issues, researchers suggest new GAN variations. For example, to solve the unstable training, non-convergence, and various modes collapses of unconditional basic GANs, the authors in [89] propose a new approach called conditional Wasserstein deep convolutional generative adversarial network (CWGAN). However, in order to evaluate their performance, these new variants must be modified for data imputation.

VI. CONCLUSION

Imputation of missing values has been one of the important challenges in recent years, and many researchers have tried to develop models for their desired problems. In this survey, we studied the common imputation algorithms containing the statistical and machine learning ones. Particularly, we focused on the GAN-based imputation algorithms, which are a class of deep learning techniques. The power of

GAN-based algorithms for generating synthetic samples has already been proved, and many researchers have contributed to designing novel GAN-based algorithms for generating synthetic data. Also, many survey papers have covered the GAN-based algorithms for synthetic data generation. However, there is no comprehensive survey for the GAN-based imputation algorithms. This paper covers 14 GAN-based imputation algorithms, which can give an idea to readers how to develop a GAN-based algorithm for a desired problem.

A. TAKE-HOME LESSONS

GAN-based algorithms, as a subset of machine learning algorithms, are adaptable for different applications. It is the reason that they have been utilized for a wide variety of applications in the context of imputation including the imputing of time-series data and missing pixels in images. Our investigation shows that there is an increasing research interest in GAN-based imputation algorithms. We have categorized the GAN-based imputation algorithms into four different groups: regular, specific, time-series, and image-based algorithms. Since different categories have different purposes, they cannot be compared to each other. However, we can convert the data to be usable in other categories. For example, TSDIGAN converts the time-series data to images and uses temporal dependency instead of spatial dependency of time-series data. Apart from the type of category, each GAN-based imputation algorithm may have some benefits and drawbacks. For instance, BiGAN can handle different lengths of time-series data and benefits from a joint training strategy for imputation and prediction. However, its training phase is difficult and suffers from the gradient vanishing problem. There are high opportunities and research gaps that can be filled in the future. The previous GAN-based algorithm can be improved, or the researchers can introduce new ones.

B. FUTURE WORKS

Most of the studies have tried to take the ability of the generator component to impute the missing values, However, they have paid less attention to the ability of the discriminator. The discriminator has many more applications such as anomaly detection [137] and also is used to generate the class labels in the form of its multi-task structure (e.g., auxiliary classifier GAN [138]). Among the investigated GAN-based algorithms, only one of them (HexaGAN) has tried to impute the missing labels beside the input features. Nevertheless, the other developed GAN-based models such as auxiliary classifier GAN can be utilized for the imputation of missing labels beside the input features. Also, researchers can convert one type of imputation category to the other types. For instance, HexaGAN has been designed for the imputation of regular data, while it can be converted to a time series or an image-based imputation algorithm by using RNN and convolutional components. There are many new GAN-based algorithms that are designed for synthetic data generation, while most of them have not been utilized for the imputation of missing data. Researchers can develop them for imputation purposes. A universal methodology for benchmarking GAN-based missing data imputation algorithms remains a work in progress. This approach is expected to accommodate varied data fractions in MCAR and MNAR situations while addressing a gap in the literature on MAR simulations. To summarize, researchers now have a lot more opportunities to use generative networks for missing data imputation.

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